11 — Glossary: Words That Appear in Problem Statements

Misreading ONE word is the most common way to fail a question. Skim this before the exam.

Words about position and order

Word	Means	Example
index / **position**	where an item sits, counting from **0**	in `[7,8,9]`, index 1 holds 8
element / **item**	one thing inside a list	8 is an element of `[7,8,9]`
adjacent / **neighbouring** / **consecutive**	next to each other	in `[1,2,3]`, 1 and 2 are adjacent
contiguous	in an unbroken run, no gaps	`[2,3]` is contiguous in `[1,2,3,4]`; `[1,3]` is not
ascending	smallest → largest	`[1,2,3]`
descending	largest → smallest	`[3,2,1]`
in order / **preserving order**	keep the original arrangement	don't sort it
in-place	change the original, don't build a new one	`nums.sort()`
reverse	backwards	`"abc"` → `"cba"`
rotate	shift items around, wrapping	`[1,2,3]` rotated left by 1 → `[2,3,1]`

Words about parts of things

Word	Means	Example
substring	a **contiguous** piece of a string	`"ell"` is a substring of `"hello"`
subarray / **sublist**	a **contiguous** piece of a list	`[2,3]` is a subarray of `[1,2,3,4]`
subsequence	items in order but **gaps allowed**	`[1,3]` is a subsequence of `[1,2,3]`
prefix	the start of something	`"he"` is a prefix of `"hello"`
suffix	the end of something	`"lo"` is a suffix of `"hello"`
digit	one numeral of a number	1234 has digits 1, 2, 3, 4
character / **char**	one letter/symbol of a string	`"a"` in `"cat"`
row	one horizontal line of a grid	`grid[0]`
column	one vertical line of a grid	gathered with a loop

> **"contiguous" is the big one.** It means no gaps. If a problem says "contiguous subarray", `[1,3]` from `[1,2,3]` does NOT count — you skipped the 2.

Words about counting and uniqueness

Word	Means
distinct / **unique**	different values, each counted once. `[1,1,2]` has 2 distinct values
duplicate	a value appearing more than once
frequency / **occurrences**	how many times something appears
most frequent / **mode**	the value appearing most often
at most	≤ (this many or fewer)
at least	≥ (this many or more)
strictly greater	> (bigger, NOT equal)
non-negative	0 or positive (0 counts!)
positive	greater than 0 (0 does NOT count)

> **"at most 2" includes 2.** "Strictly more than 2" does not. These decide whether you write `<=` or `<`.

Word puzzles that show up constantly

Word	Means	Example
palindrome	reads the same forwards and backwards	`"level"`, `"radar"`, `12321`
anagram	same letters, rearranged	`"listen"` and `"silent"`
isogram	no repeated letters	`"cat"` yes, `"cattle"` no
vowel	a, e, i, o, u	everything else is a **consonant**
alphanumeric	letters and digits only (no spaces/punctuation)	`.isalnum()`
case-insensitive	treat A and a as the same	use `.lower()` on both sides
whitespace	spaces, tabs, newlines	`.strip()` removes it from the ends

How to check each:

- palindrome → `s == s[::-1]`
- anagram → `sorted(a) == sorted(b)`
- isogram → `len(set(s)) == len(s)`

Words about the function itself

Word	Means
parameter / argument	the value handed to your function (`def f(nums)` — `nums` is the parameter)
return	hand a value back. **The grader reads this — `print` scores zero**
predicate	a function returning True/False
iterate / traverse	loop over
implement	write the code for
modify	change
preserve	keep unchanged
edge case	an unusual input: empty, one item, all-same, biggest allowed
constraint	a promise about the input, e.g. `1 ≤ n ≤ 1000`
time limit exceeded	your code is too slow — needs a better approach

> **Constraints are a hint.** `n ≤ 1000` means a double loop is fine. `n ≤ 100000` means you need a set/dict or a single pass.

Words about grids

Word	Means
matrix / grid / 2D array	a list of lists
cell	one position, `grid[r][c]`
square matrix	same number of rows and columns
diagonal	top-left → bottom-right (`grid[i][i]`)
anti-diagonal	top-right → bottom-left
transpose	swap rows and columns
boundary / border / edge	the outer ring of cells
8-directionally adjacent	including diagonals (Minesweeper)
4-directionally adjacent	up/down/left/right only

Maths words

Word	Means
sum	added together
product	multiplied together
remainder / modulo	what's left after division — `%`

Word	Means
quotient	the result of division
divisible by k	`n % k == 0`
factor / **divisor**	a number that divides evenly
multiple	6 is a multiple of 3
prime	only divisible by 1 and itself
absolute value	distance from zero, always positive — `abs()`
average / **mean**	`sum(nums) / len(nums)`
integer	whole number
inclusive	include the endpoints. "1 to n inclusive" → `range(1, n+1)`
exclusive	don't include the endpoint

Phrases that tell you the ANSWER'S SHAPE

Read these carefully — they decide what you return.

The problem says	You return
"how many" / "count" / "the number of"	a **number**
"return the sum/total"	a **number**
"return all ___" / "return the ___s"	a **list**
"return the first ___"	a **single item**
"return true if" / "determine whether"	**True/False**
"return the index of"	a **number** (a position, not the value!)
"return a new ___"	build fresh, don't modify the input

> The worked example shows you the shape. `[1,2,3] -> 2` is a number. `[1,2,3] -> [2]` is a list. Look at the brackets.